CSE 4501 (SWE): OS - Quiz #4 **20 March 2025**
Duration: 20 Minutes **Full Marks: 15**
Student ID:

1. Consider a logical address space of 64 pages of 1,024 words each, mapped onto a physical memory of 32 frames. $2 + 2$

   a) How many bits are there in the logical address?

   b) How many bits are there in the logical address?

Solution:

LAS - 64 pages
Size of each Page - 1024
PAS - 32 frames
Size of each frame - 1024

Bits in Logical Address space $= log_2(64) + log_2(1024) = 6 + 10 = 16$ bits
Bits in Physical Address space $= log_2(32) + log_2(1024) = 5 + 10 = 15$ bits

2. Suppose a system uses a Logical address space of 32 bits, with a page and frame size of 4 KB. The system uses two-level paging, where the first-level page table has 16 entries, and the second-level page table has 1024 entries. Assuming that all second-level page tables have identical entries, convert the Logical Address (LA) `(2 * x + 1000)` to the Physical Address (PA) using two-level paging. Calculate the First Page Index, Second Page Index, and the offset from the given LA to convert it to the PA. Also, calculate the base address of the second-level page table in converting to the PA. Remember that 11

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x = last five digits of Student ID
y = last four digits of Student ID
z = last two digits of Student ID
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Table 1: First Page Table

First-Level Index (FPI)	Base Address of Second-level table
0	y + (FPI*4096)
1	y + (FPI*4096)
2	y + (FPI*4096)
...	...
...	...
13	y + (FPI*4096)
14	y + (FPI*4096)
15	y + (FPI*4096)

Table 2: Second Page Table

Second-Level Index (SPI)	Frame Number (FN)
0	SPI*((z mod 4) + 1)
1	SPI*((z mod 4) + 1)
...	...
592	SPI*((z mod 4) + 1)
593	SPI*((z mod 4) + 1)
...	...
1022	SPI*((z mod 4) + 1)
1023	SPI*((z mod 4) + 1)

Solution:

Assuming student id = 210042157.
x = 42157.
y = 2157.
z = 57.

LA = 2 * 42157 + 1000.
$= (85314)_{10} = (0000\ 0000\ 0000\ 0000\ 0101\ 0011\ 0001\ 0010)_2$

10 bits for the first-level page index.

10 bits for the Second-level page index.
12 bits for the offset.

first-level page index = $(0000\ 0000\ 00)_2 = (0)_{10}$
Second-level page index = $(00\ 0000\ 0101)_2 = (5)_{10}$
Offset = $(0011\ 0001\ 0010)_2 = (170)_{10}$

The base address for the second-level page table at the 0th index = 2157 + (0 * 4096) = 2157
Frame number from second-level page table at the 5th index = 5*((57 mod 4) + 1) = 10

PA = Frame number * page size + offset = 10 * 4096 + 170 = 41130

Metrics:

- Calculating LA - 1 mark
- Calculating FPI from LA - 2 mark
- Calculating Base address from FPI - 1 mark
- Calculating SPI from LA - 2 marks
- Extracting Frame Number from Second Page Table - 1 mark
- Calculating Offset - 2 mark
- Converting LA to PA - 2 marks